INCH-POUND

MIL-R-23417/29A
25 April 1991

SUPERSEDING
MIL-R-23417/29
24 October 1966

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 11R20W4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Tachometer Generators described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements.

Requirements	Value	Unit	Tolerance
Excitation winding	Stator		
Rated frequency	400	Hz	±2%
Input voltage:			
Maximum	26	volts (rms)	±1%
Test	10	volts (rms)	±1%
Minimum	0.5	volts (rms)	±1%
Primary current	6.0	milliamps	maximum
Primary power	0.01	watts	nominal
D.C. resistance:			
Rotor	320	ohms	±15%
Stator	305	ohms	±15%
Compensator	370	ohms	±15%
Impedance:			
Zro	420+j2040	ohms	R±15%+jx±10%
Zso	410+j1920	ohms	R±15%+jx±10%
Zrs	610+j230	ohms	R±15%+jx±10%
Zco	470+j1910	ohms	R±15%+jx±10%
Resonant frequency:			
Rotor	70	kHz	minimum
Compensator	80	kHz	minimum
Low corner frequency	75	Hz	maximum

AMSC - N/A

FSC 5990

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TABLE I. Requirements - Continued.

Requirements	Value	Unit	Tolerance
Transformation ratio:			
Output/input	0.980	----	±0.010
Rotor/compensator	0.990	----	±0.010
T. R. variation (R/C) due to voltage (with test voltage as reference):			
@1-10v	±0.10	percent	maximum
@11-26v	±0.05	percent	maximum
Equality of transformation ratio:			
Output/input	0.1	percent	maximum
Rotor/compensator	0.1	percent	maximum
Phase shift:			
Output/input	9.0	degrees	±1.2
Rotor/compensator	0	degrees	±0.10
Phase shift variation (R/C) due to voltage:			
@1-26v	±4.0	degrees	maximum
Interaxis error	±3.0	minutes	maximum
Functional error	±0.1	percent	maximum
Null voltage:			
Fundamental	1.0	mv/v output	maximum
Total	1.5	mv/v output	maximum
Shift of resolver zero:			
With voltage	±1.5	minutes	maximum
With frequency	±0.50	minutes	maximum
Harmonic distortion	0.02	percent	maximum
Temperature rise	5.0	degrees C	maximum
Mechanical breakaway torque:	0.1	oz-in	maximum
@ -55°C	0.2	oz-in	maximum
Radial play	0.0006	inch	maximum
End play	0.0010	inch	±0.0005

NOTES:

1. A voltage value of 0.5 to 26 volts shall be entered in the "volts" designation space on the identification plate.

2. Transformation ratio variation with input voltage will be tested in accordance with MIL-R-23417 with measurements made at 1, 5, 10, 15, 20, and 26 volts.

3. This resolver can be used for all types of trigonometric problem solving, precise data transmission data, and in computing chains. (See MIL-HDBK-218, Application of Electrical Resolvers.) External trimming can be applied for desired Resolver-Amplifier system requirements.

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TABLE II. Part identifying number variant characteristics.

Part Identifying No. 1/	A ±.010	B ±.010	G Maximum	Shaft Style	Terminal End	U ±.0000 -.0005
M23417/29-01A	.555	.500	1.789	Spline	Terminal Board	.1872
M23417/29-02A	.367	.312		Smooth	Wire Leads	.1200
M23417/29-03A	.430	.375		Smooth	Wire Leads	.1200
M23417/29-04A	.492	.437		Smooth	Wire Leads	.1200
M23417/29-05A	.555	.500		Smooth	Wire Leads	.1200
M23417/29-06A	.617	.562		Smooth	Wire Leads	.1200
M23417/29-07A	.367	.312		Smooth	Terminal Board	.1200
M23417/29-08A	.430	.375		Smooth	Terminal Board	.1200
M23417/29-09A	.492	.437		Smooth	Terminal Board	.1200
M23417/29-10A	.555	.500		Smooth	Terminal Board	.1200
M23417/29-11A	.617	.562	1.789	Smooth	Terminal Board	.1200

1/ Part identifying number M23417/29-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

TABLE III. Part identifying number cross-reference. 1/

SUPERSEDED PART NUMBERS	NEW PART IDENTIFYING NUMBERS
MIL-R-23417/29	
M23417/29-01	M23417/29-01A
M23417/29-02	M23417/29-02A
M23417/29-03	M23417/29-03A
M23417/29-04	M23417/29-04A
M23417/29-05	M23417/29-05A
M23417/29-06	M23417/29-06A
M23417/29-07	M23417/29-07A
M23417/29-08	M23417/29-08A
M23417/29-09	M23417/29-09A
M23417/29-10	M23417/29-10A
M23417/29/11	M23417/29-11A

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

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Custodians:
Army - AR
Navy - AS
Air Force - 85

Review activities:
Navy - EC
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - MI
Navy - CG, MC
Air Force - 14, 19

Preparing activity:
Army - AR

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